

Final Project

STUDENT GRADE MANAGEMENT SYSTEM

Link: <https://github.com/DefNT/Project>

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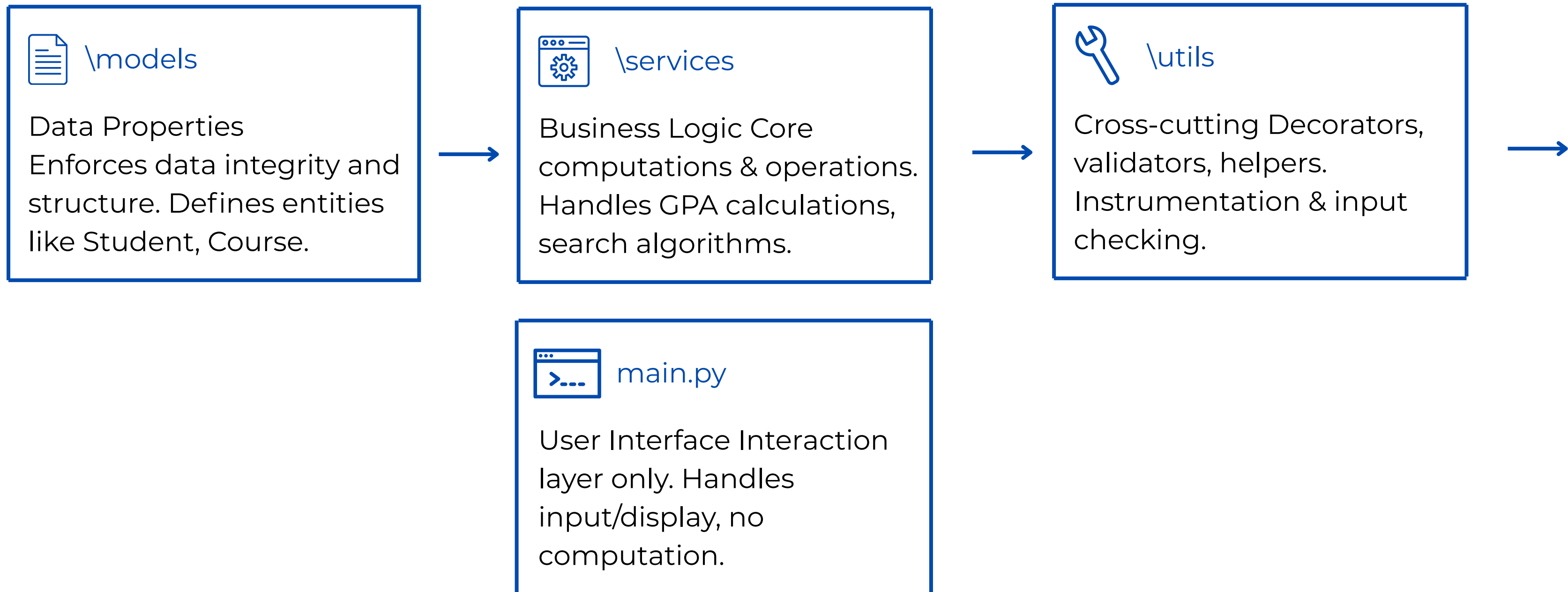
The Manual Tracking bottleneck

- ✗ Traditional spreadsheets cause data fragmentation across disconnected files and departments.
- ✗ Manual tracking leads to input errors, broken formatting, and inconsistent records.
- ✗ Inefficient reporting when processing hundreds of student records under deadline pressure.

Automated Engineering Approach

- ✓ A modular CLI application with strict validation rules ensuring data correctness at every entry point.
- ✓ Data persistence via reliable multi-file CSV tracking with atomic read/write cycles
- ✓ Automated edge-case checking (e.g., scoring bounds of 0–100) eliminates entire classes of bugs.

Architectural Separetion



OOP

Inheritance

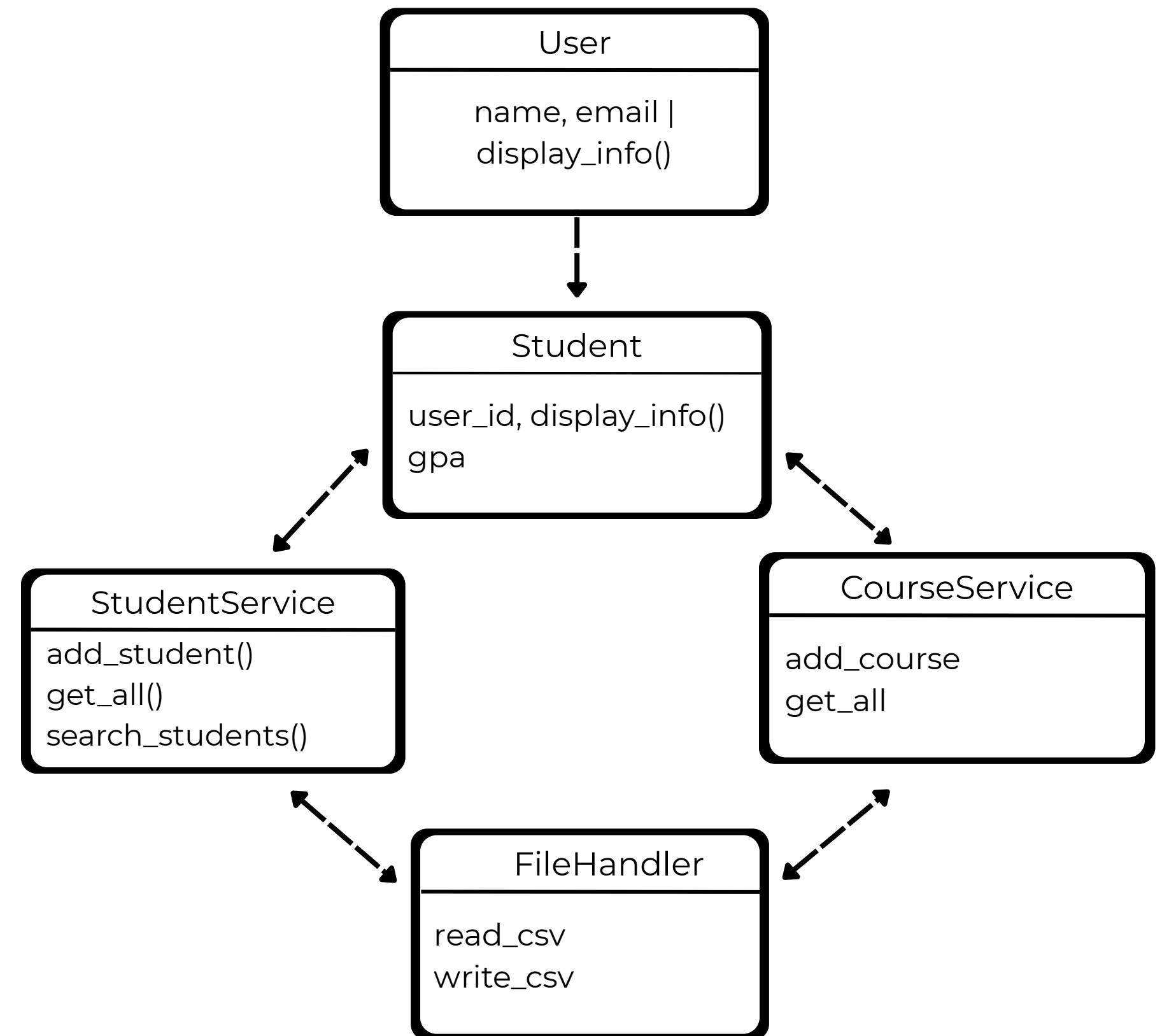
Base **User** class cleanly extended by **Student** to manage domain-specific metadata like grades, GPA, and enrollment records.

Polymorphism

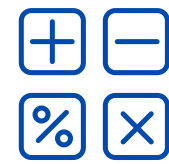
Overridden **display_info()** methods across classes for dynamic, context-aware console reporting without conditionals.

Encapsulation

Complete separation of concerns. UI controllers cannot touch raw file layers directly; all data passes through managed Service abstractions.



GPA Calculation Framework



Arithmetic GPA Engine

Processes raw mean points across all completed records.
Simple average calculation without weight consideration

$$\text{GPA} = \Sigma(\text{Scores}) / \text{Count}$$

```
def calculate_simple_gpa(self, sid): 7 usages (2 dynamic)  yerraxd
    grades = self.get_student_grades(sid)
    if not grades:
        return 0.0
    return round(sum(g.score for g in grades) / len(grades), 1)
```



Credit-Weighted Engine

Cross-references active credit valuations to compute
proper creditweighted calculations for accurate
academic standing.

$$\text{GPA} = \Sigma(\text{Score} \times \text{Credits}) / \Sigma(\text{Credits})$$

```
def calculate_weighted_gpa(self, sid): 5 usages (2 dynamic)  yerraxd +1
    grades = self.get_student_grades(sid)
    if not grades:
        return 0.0
    pts = 0.0
    crds = 0
    for g in grades:
        c = self.course_service.courses.get(g.course_id)
        crd = c.credits if c else 3
        pts += g.points * crd
        crds += crd
    return round(pts / crds, 1) if crds else 0.0
```


DEMO

Starting Menu

```
=====
      STUDENT GRADE MANAGEMENT SYSTEM
=====
loading data...
Loaded: 100 students, 8 courses

Press Enter to continue...|
```

Main UI

```
=====
      STUDENT GRADE MANAGEMENT SYSTEM
=====
Students: 100 | Courses: 8

[1] Students
[2] Courses
[3] Grades
[4] Analytics & Reports
[5] Save data
[0] Exit

Choice: |
```

Student UI and lising all students

```
=== STUDENTS ===
[1] List all students
[2] Add student
[3] Search student
[4] View student
[0] Back

Choice: 1

=== ALL STUDENTS ===

ID      Name          Major          GPA    W.GPA
-----
S1      Student_1         Physics        69.0    1.0
S2      Student_2         Mathematics    72.0    2.1
S3      Student_3         Physics        77.3    2.4
S4      Student_4         FinTech       82.6    2.9
S5      Student_5         Mathematics    87.4    3.4
```

Student UI and viewing student

```
=== STUDENTS ===
[1] List all students
[2] Add student
[3] Search student
[4] View student
[0] Back

Choice: 4

=== STUDENT REPORT ===

Student ID: S5
[22:51:03] Running: _student_report()

Name: Student_5
Major: Mathematics
Simple GPA: 87.4
Weighted GPA: 3.4

Course      Name                      Cr   Score Grade
-----
CS101      Intro to Programming      3    93.3   A
MATH201     Linear Algebra            4    80.6   B
CS301      Operating Systems         4    83.3   B
MATH101     Calculus I                4    99.7   A
CS401      Blockchain Tech           3    80.2   B
Timer: _student_report() took 0.0013 seconds

Press Enter to continue...|
```

DEMO

Courses UI and lising all courses

```
=== COURSES ===

[1] List all courses
[2] Add a course
[3] View course stats
[0] Back

Choice: 1

=== ALL COURSES ===

Code      Name                Credits Students
-----
MATH101    Calculus I             4         56
CS101      Intro to Programming    3         51
PHYS101    General Physics        4         62
CS202      Data Structures        3         49
FIN101     Intro to FinTech       3         51
CS301      Operating Systems      4         51
MATH201    Linear Algebra         4         50
CS401      Blockchain Tech        3         53

Press Enter to continue...|
```

Courses UI and course stats

```
=== COURSES ===

[1] List all courses
[2] Add a course
[3] View course stats
[0] Back

Choice: 3

=== COURSE STATS ===

Course code: CS101

Course: CS101
Students: 51
Mean: 73.2
Median: 70.3
Min: 50.0 Max: 99.9
Pass rate: 74.5%

Distribution: A:13 B:4 C:9 D:12 F:13

Press Enter to continue...|
```

Analytics UI and Top students

```
==== Analytics ====
[1] Top students
[2] Hall of Fame (Gpa>=90)
[3] Failing students
[4] Full report
[5] Export report to JSON
[6] Export grades to CSV
[0] Back

Choice: 1

=== TOP STUDENTS (by weighted GPA) ===

How many? (default 5):5

Rank  Name                Major                W.GPA
-----
1      Student_47           Software Engineering  3.7
2      Student_16           Software Engineering  3.6
3      Student_62           FinTech               3.6
4      Student_6            Physics               3.5
5      Student_5            Mathematics           3.4

Press Enter to continue...|
```

DEMO

Analytics UI & Hall Of Fame feature

```
==== Analytics ===
[1] Top students
[2] Hall of Fame (Gpa>=90)
[3] Failing students
[4] Full report
[5] Export report to JSON
[6] Export grades to CSV
[0] Back

Choice: 2

== HALL OF FAME (Weighted GPA>=90) ==

Name                Major                W.GPA
-----
Student_47          Software Engineering 3.7
Student_16          Software Engineering 3.6
Student_62          FinTech              3.6
Student_6           Physics              3.5

Press Enter to continue...|
```

Analytics UI & failing students feature

```
==== Analytics ===
[1] Top students
[2] Hall of Fame (Gpa>=90)
[3] Failing students
[4] Full report
[5] Export report to JSON
[6] Export grades to CSV
[0] Back

Choice: 3

=== FAILING STUDENTS (GPA<60) ===

Student_36 - GPA: 55.7
Student_79 - GPA: 59.1

Press Enter to continue...|
```